



Applying the IGLUE Best Practices with Escapp

Title

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Table of Contents

1. Introduction	1
2. Design guidelines	2
2.1. Defining appropriate learning objectives	2
2.2. Choosing an appropriate format and delivery mode	2
2.3. Choosing an appropriate participation mode and team size	3
2.4. Establishing an appropriate duration	3
2.5. Crafting a consistent and engaging narrative	4
2.6. Structuring the puzzles appropriately	5
2.7. Designing appropriate puzzles	7
2.8. Establishing an appropriate difficulty	10
2.9. Defining how to provide participants with timely help	11
2.10. Defining the strategy of student grading	12
2.11. Designing the escape room with accessibility and inclusivity in mind	12
3. Build guidelines	14
3.1. Building the escape room with reusability in mind	14
3.2. Using resources with low environmental impact	15
3.3. Considering changing the design of the escape room to make it easier to build	15
4. Execution guidelines	16
4.1. Considering scheduling shifts and allowing asynchronous participation	16
4.2. Providing participants with initial instructions	16
4.3. Preparing the escape room for its execution	17
4.4. Supervising the escape room whenever feasible	17
4.5. Monitoring participants while playing the escape room	19
4.6. Helping participants to solve the puzzles when necessary	20
4.7. Motivating participants while playing the escape room	21
4.8. Conducting a debriefing session	22
5. Evaluation guidelines	23
5.1. Testing the escape room thoroughly before its execution	23
5.2. Testing the escape room from the early stages of development	24
5.3. Evaluating the impact of the escape room on participants' perceptions and learning	24

1. Introduction

Within the scope of the IGLUE project, we previously developed the guide titled "**Best Practices for Creating and Conducting Educational Escape Rooms: A Comprehensive Guide for Teachers**" (or "**IGLUE Best Practices**" for short), which provides technology-agnostic guidelines and recommendations to help teachers of any educational level and knowledge area to design, build, execute, and evaluate educational escape rooms considering different formats (physical, virtual, and hybrid) and delivery modes (face-to-face and remote).

Within the IGLUE project, we also developed **Escapp**, a virtual platform for creating, conducting, and sharing educational escape rooms. The **Escapp user manual** explains how to use the main features of this platform.

The present document, titled "**Applying the IGLUE Best Practices with Escapp**", provides considerations, instructions, and recommendations for teachers on how to apply the IGLUE Best Practices when using Escapp. This document does not replace the IGLUE Best Practices guide, but it extends it by providing additional information and recommendations on how to apply these practices when a specific tool is employed. The target audience of the present document is teachers of any educational level and knowledge area who are interested in creating and conducting educational escape rooms using Escapp.

The present document is organized according to the guidelines defined in the IGLUE Best Practices guide and the four phases of the educational escape room life cycle proposed in this guide: design, build, execution, and evaluation. There is a section for each of these phases and, within each of these sections, a subsection for each guideline.

2. Design guidelines

2.1. Defining appropriate learning objectives

Escapp supports the definition of **learning objectives** for an educational escape room through its **Basic Configuration** section, which is the first section teachers encounter when creating a new escape room. While Escapp does not have a dedicated field specifically labelled "learning objectives", this section provides a **description** field that teachers can use to specify the learning objectives of an educational escape room with the desired level of detail. Moreover, this section also provides specific fields for indicating the **educational level** at which the escape room is targeted, the **knowledge area** covered, and the specific **topics** addressed. This information can add further value to the definition of the learning objectives.

Another section that teachers can use to specify the learning objectives of an educational escape room is the **Pre-activity instructions** section. In this section, teachers can write out the objectives using a rich text editor, and even attach supplementary documents (e.g., a PDF describing the objectives in detail). In this regard, it should be pointed out that users will only be able to see the **pre-activity instructions** after enrolling in the escape room.

2.2. Choosing an appropriate format and delivery mode

Escapp can be used to create **virtual and hybrid educational escape rooms**, as well as to conduct these escape rooms both **face-to-face and remotely**. Naturally, since Escapp is a virtual platform, by definition it cannot be used to conduct a physical escape room, as introducing Escapp into a physical escape room would make it a hybrid one.

In the **Basic Configuration** section, teachers can indicate whether the **format** of an educational escape room is **virtual or hybrid**. If the hybrid format is selected, Escapp enables the escape room's author to provide a downloadable file with additional resources (e.g., printable materials) in the **Pre-activity instructions** section. Thus, teachers can provide the physical resources needed to play the escape room through this downloadable file or in any other way outside Escapp (e.g., distributing physical components in the classroom).

Regarding the **delivery mode**, Escapp supports both **face-to-face and remote** participation. For **face-to-face** delivery, teachers can configure a screen from the **Classroom interface** section and project that screen in the classroom in order to display the same content to all participants. This content can include, for example, an introductory video, a countdown timer or a leaderboard. When using an introductory video in escape rooms delivered face-to-face, we recommend displaying the video using the **classroom interface** instead of the **team interface** whenever possible, provided that all participants can see and hear the video properly. For **remote** delivery, teachers can provide participants with a **link to a video conferencing room** (configured in the **Basic configuration** section) where they can seek help or resolve technical issues during the activity.

Lastly, it is worth noting that Escapp allows teachers to create **multiple shifts** for an educational escape room from the **Shifts** section and offer both face-to-face and remote shifts for the same escape room.

2.3. Choosing an appropriate participation mode and team size

Escapp allows teachers to specify the **participation mode** and **team size** for an educational escape room in the **Basic Configuration** section using two fields: **maximum team size** and **minimum team size**. In order to specify that an escape room **allows only individual participation** (i.e., that participants can only play individually), teachers **must enter a value of 1 in the maximum team size field**. Any other value indicates to Escapp that the escape room can be played in teams. When an escape room allows only individual participation, the team selection step is skipped when enrolling.

When an escape room allows participation in teams, Escapp allows participants to form new teams and join existing teams during enrollment. If the **maximum team size** field is left blank, Escapp will allow teams of any size. However, if a number is specified in this field, Escapp will restrict the maximum number of members a team can have to the specified value. Escapp also provides teachers with a **minimum team size** field to specify the minimum number of members a team should have to participate in a specific escape room. However, it should be taken into account that the value specified in this field is for informational purposes only, since Escapp will not use it to impose any restriction during team formation. Therefore, when a teacher wants to enforce a specific minimum team size, it is recommended to communicate that requirement to participants as part of the activity rules (e.g., by using the **Pre-activity instructions** section).

Teachers can see the participants and teams enrolled in an escape room from the **Participants** and **Teams** sections. Teachers can also perform several actions for each team or, when individual participation mode is enabled, for each participant, such as restarting their progress, deleting their enrollment, sending an email, or sending a real-time message.

2.4. Establishing an appropriate duration

Escapp allows teachers to specify the **duration** in minutes of an educational escape room in the **Basic Configuration** section using a specific field called **duration**. The value specified in this field will be used for informational purposes, as well as in the countdown timers displayed to participants during the escape room to indicate how much time they have left. Furthermore, teachers can choose whether the value specified in the **duration** field should be used as a strict time limit, in which case Escapp will prevent participants from solving the escape room puzzles once the time limit has expired.

In addition to specifying the duration of the whole escape room, Escapp also allows teachers to specify the expected duration in minutes for each escape room puzzle in the **Puzzles and hints** section. Thereby, teachers can specify the amount of time they estimate participants will require to solve each puzzle. These durations can be used by Escapp to automatically and proactively deliver hints or suggestions to request hints to participants.

Lastly, it should be noted that the time specified in the **duration** field in Escapp corresponds specifically to the duration of the educational escape room, excluding any complementary activities intended to be performed alongside it, such as introductory sessions, pre-tests and post-tests, and debriefing sessions. The duration of these complementary activities, as well as the total duration of the experience (including the educational escape room and its complementary activities), can be specified using the **Pre-activity instructions** section, the **Post-activity instructions** section, and/or the **description** field.

2.5. Crafting a consistent and engaging narrative

The **narrative** of an educational escape room refers to its specific story, including its theme or setting, the role of the participants, their allies, the conflicts, and the final goal. The IGLUE Best Practices guide includes various recommendations aimed at making the narrative of an escape room consistent and engaging. This section provides information that may help teachers follow these recommendations when creating an escape room using Escapp.

A good way of presenting the narrative of an educational escape room to its participants is to use an **introductory video**. An introductory video can be an excellent way of **clearly presenting the storyline, setting, context, participants' roles, and final goal of an escape room**, as well as **giving participants compelling reasons why they should accomplish this goal and why they should do so within the given time frame**. An introductory video can also help to **start with a strong hook that captures participants' attention from the beginning**.

Teachers have different ways of providing an introductory video in an escape room created with Escapp. Escapp allows teachers to upload videos and add them to the **resource catalog** of an escape room. These videos can later be included in the **pre-activity instructions**, allowing participants to watch them from their own devices before starting the escape room. Videos can also be included in the **team interface**, allowing participants to watch them from their own devices during the escape room. It is possible to include a video for participants to watch immediately after beginning to play the escape room. When including a video in the **team interface**, teachers can opt to include the video with a standard video player in a content block or by using an **interactive resource offered in the Escapp's resource catalog**, such as a TV or a custom player. This latter option allows teachers to require participants to watch the whole video before advancing in the escape room. Moreover, teachers can also include videos in the **classroom interface**, which allows them to show the videos to a whole class or group of participants at the same time. When using an introductory video in escape rooms delivered face-to-face, we recommend displaying the video using the **classroom interface** instead of the **team interface** whenever possible, provided that all participants can see and hear the video properly. When the introductory video is displayed through the **classroom interface**, the escape room can be initiated manually by the teacher immediately after the video ends.

When using Escapp, teachers can also present the narrative of an educational escape room in other ways, such as through **audio** (e.g., simulating a phone call), **images**, and/or **enriched text content**. Escapp allows teachers to upload audio files and images and include them in escape rooms, as well as to create content using a rich text editor, which allows teachers to combine text and images.

The IGLUE Best Practices guide recommends **maintaining coherence across all elements of an escape room**. Escapp has two features that can help teachers follow this recommendation. Firstly, several of the **interactive resources that Escapp offers through its resource catalog** allow teachers to choose between different **skins**, enabling them to customize their appearance to match the escape room theme. For example, for an educational escape room set in a medieval town, a teacher could add a "Switches" resource with the "Retro" skin so that participants interact with medieval torches instead of modern electronic switches, which would not be consistent with the escape room's medieval theme. Secondly, Escapp offers different **visual themes** that allow teachers to change the overall appearance of the **team interface**, adjusting various aspects such as the background and the default text color. This feature can help make the interface seen by participants better match the escape room narrative (e.g., by choosing a dark theme for a mystery story).

Escapp provides a tool called **Scene Maker** that can help teachers **create strong and vivid settings for educational escape rooms**. This tool allows teachers to create interactive 2D scenes for educational escape rooms in which participants can explore a series of screens, interact with characters and objects, and solve puzzles, in a similar way to classic point-and-click adventure games. This tool can also be helpful for **incorporating and developing characters** in escape rooms who may offer clues or background information, pose challenges, or act as distractors.

Escapp allows teachers to display different content to the participants of an escape room as they progress through it, which may be very helpful for **incorporating mystery, suspense, surprises, and/or unexpected twists into the narrative**. The **content blocks** of the **team interface** can be unveiled progressively to participants as they solve the **escape room puzzles**. Teachers can specify, for each content block, the escape room puzzles during which the content should be displayed to participants. In this way, teachers can unfold the narrative progressively, revealing new story elements as each puzzle is solved. In addition to content blocks, teachers can define **success messages** for the puzzles, which may also be helpful for advancing the narrative.

To **provide a satisfying conclusion** to an escape room created with Escapp, teachers can add content blocks to the **team interface** that are displayed to participants only when they complete the escape room. Thus, teachers can use these blocks to deliver the final content and provide the desired conclusion.

2.6. Structuring the puzzles appropriately

The IGLUE Best Practices guide indicates that the structure of puzzles in an escape room can be mainly of two types: **linear** and **non-linear**. In an escape room with a linear puzzle structure, the puzzles are arranged in a sequence in such a way that each puzzle cannot be completed until the previous ones have been solved, which forces participants to solve the escape room puzzles in a specific order. In an escape room with a non-linear puzzle structure, there is no single order in which participants must solve the puzzles, but, at certain times, there are multiple puzzles that participants can solve without following a specific order. The IGLUE Best Practices guide also describes the main advantages and disadvantages of each of these types of puzzle structures.

Escapp **only supports linear puzzle structures**, which implies that:

- When teachers create the puzzles of an educational escape room from the **Puzzles and hints** section provided by Escapp, the puzzles are always arranged in a specific sequence. Therefore, each of these puzzles always has a specific position in the escape room. For example, an escape room with two puzzles will always have a Puzzle 1 followed by a Puzzle 2.
- When participants solve one escape room puzzle created through Escapp, they automatically advance to the next puzzle created through Escapp, unless the solved puzzle is the last one, in which case they successfully complete the escape room.
- At any moment from when a participant starts playing an escape room until the escape room ends, there is one (and only one) puzzle created through Escapp that the participant is facing. In Escapp, this puzzle determines the participant's current progress and the puzzle for which hints are delivered.

Although Escapp only supports linear puzzle structures, **it is possible to use Escapp to conduct educational escape rooms with a non-linear puzzle structure with some limitations.** The puzzles created for an educational escape room in Escapp will always be arranged in sequence and, from the point of view of Escapp, all its escape rooms will always have a linear puzzle structure. However, it is possible to create an escape room puzzle in Escapp whose solving requires participants to solve, in no specific order, multiple subpuzzles that have not been created through Escapp, which makes the actual puzzle structure of the escape room non-linear. For instance, as illustrated in Figure 1, it is possible to create an escape room in Escapp with 3 puzzles arranged in sequence (Puzzles 1, 2, and 3), but that actually has 5 puzzles following a non-linear structure, since solving Puzzle 2 requires solving Subpuzzles 2A, 2B, and 2C. Puzzle 2 of this escape room should be created in Escapp in such a way that participants can obtain its solution from the solutions of its subpuzzles (e.g., by concatenating these solutions or performing mathematical operations on them).

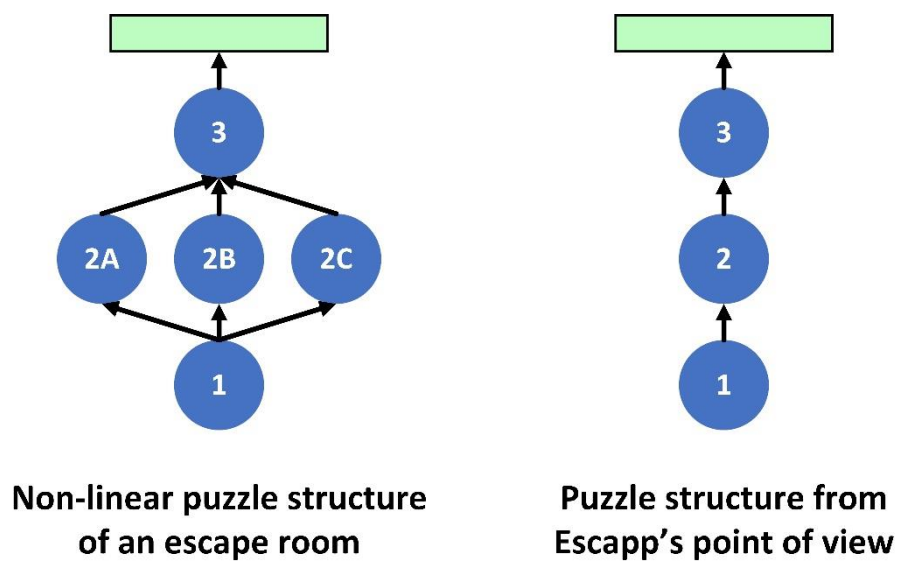


Figure 1: Escape room with a non-linear puzzle structure from Escapp's point of view

Hereafter, for simplicity, we will refer to the escape room puzzles that have not been created through Escapp as non-Escapp puzzles. Although non-Escapp puzzles are not managed by Escapp, teachers can define **hint categories** in order to allow participants to request hints for any of them. For the escape room represented in Figure 1, teachers can define hints for Subpuzzles 2A, 2B, and 2C and allow participants to request them through Escapp during Puzzle 2.

It should be taken into account that conducting educational escape rooms with a non-linear puzzle structure with Escapp requires the use of non-Escapp puzzles and hence has significant limitations:

- Escapp cannot verify solutions for non-Escapp puzzles and thus participants cannot submit solutions for these puzzles through Escapp. For the escape room represented in Figure 1, Escapp will only verify solutions for Puzzle 2, but not for Subpuzzles 2A, 2B, and 2C.
- Escapp is not aware when an individual non-Escapp puzzle is solved and thus, when one of these puzzles is solved, it cannot update the escape room state, the content shown to the participants, or their progress or grade.

- The resources offered by the Escapp's resource catalog cannot be linked to non-Escapp puzzles and thus these puzzles cannot be solved by interacting with these resources.
- Proactive delivery of hints cannot rely on the expected duration of an individual non-Escapp puzzle.
- Escapp will provide less data on participants' interactions, since it will not record when non-Escapp puzzles are solved or the failed attempts to solve these puzzles.

2.7. Designing appropriate puzzles

The IGLUE Best Practices guide describes various requirements whose fulfillment is highly recommended for all puzzles in an educational escape room. This section provides information aimed at helping teachers create puzzles with Escapp that meet these requirements.

Requirement 1: The solution to the puzzle should be unique and easily verifiable in an objective way

Escapp allows teachers to create **puzzles with unique solutions for which the participants' answers can be automatically verified**. A specific solution for each escape room puzzle can be defined from the **Puzzles and hints** section using the **solution** field. From this section, teachers can also choose the method that Escapp will use to verify participants' answers from the following four options: **exact verification**, **relaxed verification**, **numeric verification**, and **regular expression verification**. If **exact verification** is selected, only answers whose value exactly matches the solution will be considered correct. **Relaxed verification** works in the same way as exact verification, except that capitalization and accents are ignored. If **numeric verification** is selected, the puzzle solution must be an integer or decimal number, and teachers can also specify a **tolerance**. If a tolerance is specified, answers whose value falls within the resulting range will be considered correct. For example, if the solution is 10 and the tolerance is 2, any answer providing a number between 8 and 12 will be considered correct. Lastly, **regular expression verification** makes it possible to check whether participants' answers match a specific pattern or format. If this verification method is selected, the regular expression must be specified in the puzzle's **solution** field. The exact verification is the best choice for puzzles with unique non-numeric solutions. The relaxed verification can be used when capitalization and accents of the puzzle solution are not important (e.g., for words which are frequently written with either an uppercase or a lowercase initial letter). The numeric verification is the best choice for puzzles whose solution is a number. The tolerance setting can be set to 0 to make the solution unique. Lastly, the regular expression verification is intended to allow checking the patterns or formats of participants' answers rather than the specific value.

In the **Puzzles and hints** section, teachers can also choose how participants must solve each puzzle: by **entering a value in a text box located at the header of the team interface** or by **interacting with a resource**, which can be a resource from the Escapp's resource catalog (e.g., a keypad, a decoder disk, or a safe box) or an external resource that communicates with Escapp via its API.

Requirement 2: The participants must be able to find the solution to the puzzle within the allotted time using the available resources.

Escapp allows teachers to provide resources progressively to participants through the **team interface** as they solve the escape room puzzles. Thereby, teachers can control the resources to which participants have access at any point in the escape room. Moreover, teachers can allow participants to obtain **hints** that help them solve any of the escape room puzzles.

Requirement 3: The probability that participants will solve the puzzle through trial and error should be very small

Escapp allows teachers to define puzzles with long solutions, enabling them to create puzzles that are very unlikely to be solved by chance or through trial and error. Moreover, the resources offered by the Escapp's resource catalog that allow participants to solve puzzles by interacting with them provide different settings that allow teachers to adjust the solution length or the number of required interactions.

Requirement 4: The puzzle should not have irreversible states so that participants, regardless of their past actions, should always have the opportunity to solve it.

In Escapp, when a participant of an escape room fails to solve a puzzle (either by submitting a wrong answer through the text box or by interacting with a resource from the Escapp's resource catalog), that failure never prevents the participant from continuing to try to solve the puzzle in the future or to successfully complete the escape room. Nonetheless, on some occasions, participants may have to wait some time before trying to solve a puzzle again. Moreover, the interactive resources offered in the Escapp's resource catalog that can be linked to puzzles never reach a state that prevents participants from solving a linked puzzle. The only case in which Escapp prevents participants from solving a puzzle and completing an escape room is when the available time runs out in an escape room that has been configured to use the duration as a strict time limit. In summary, teachers do not need to perform any specific action to guarantee the fulfillment of this requirement.

Requirement 5: Participants should be clear about when they have solved the puzzle.

In Escapp, whenever a participant solves an escape room puzzle, a **success message** indicating this milestone is displayed through the **team interface** to that participant and to their teammates in case they are playing in a team. Furthermore, participants can see at any point the puzzles they have solved by clicking the **information** button available in the header of the **team interface**. Moreover, whenever a participant enters a wrong answer for an escape room puzzle through the text box available in the **team interface**, a **failure message** indicating this failure is displayed through this interface. Escapp allows teachers to customize the content of these success and failure messages from the **Puzzles and hints** section. If no custom values are defined, Escapp will display default messages.

The **resources offered by the Escapp's resource catalog** that allow participants to solve puzzles by interacting with them can provide feedback to participants indicating that they have solved the puzzles. This feedback can be of different types depending on the specific resource (e.g., it can be textual, graphical, auditory, or audiovisual), and some resources allow teachers to customize it through different settings. Generally, the resources offered by the Escapp's resource catalog that can be linked to puzzles can also provide feedback after failed attempts to solve the linked puzzles.

Requirement 6: The puzzle should have a reason to exist consistent with the narrative.

Section 2.5 describes Escapp features that can help teachers make puzzles consistent with the narrative.

Requirement 7: The puzzle should be connected or related to other escape room puzzles.

Escapp allows teachers to create educational escape rooms in which puzzles are arranged in sequence. The use of this **linear puzzle structure** facilitates the connection of the escape room puzzles in a logical way. Furthermore, Escapp allows teachers to display new **content blocks** to participants and hide previously displayed ones each time they solve a new puzzle. Teachers can use these content blocks to establish relationships among the different escape room puzzles and structure the entire experience. The **success messages** displayed to participants after solving a puzzle can also be helpful for connecting the escape room puzzles.

Requirement 8: The puzzle should be tied to specific educational goals and combine educational and playful elements.

Escapp provides teachers with a **resource catalog** that offers various **configurable resources that can be linked to escape room puzzles** so that participants can only solve these puzzles by interacting with these resources. These resources can be customized through different settings and can be very helpful to teachers in creating puzzles that combine educational and playful elements. Escapp provides instructions for all these resources, which describe all available settings.

Next, we include a table that lists some of the resources available in the Escapp's resource catalog and describes possible use cases for them.

Resource name	Use cases
Keypad	Puzzles whose solution is a sequence of numbers, letters, colors or symbols.
Decoder disk	Puzzles whose solution is a sequence of up to five numbers, letters, colors, or symbols.
Safe box	Puzzles whose solution is a sequence of positive or negative numbers or angles.
Switches	Puzzles whose solution consists of a sequence of up to 14 Boolean values (e.g., true/false values).
Wire connection	Puzzles whose solution consists of up to 14 matches between two elements identified or represented by colors, numbers, letters, or symbols. This resource allows teachers to present matching questions in a gamified way.
Hidden path	Puzzles whose solution consists of a path or a set of coordinates on a two-dimensional grid.
Basic text box	Puzzles whose solution is a value of any type (e.g., a word, a number, or a combination thereof) that participants must enter through a text box. This

	resource can be used to allow participants to solve puzzles directly from a scene by using a customized text box, eliminating the need to resort to the text box provided in the header of the team interface. In this way, Escapp enables the creation of educational escape rooms that take place entirely within a scene. This resource also allows teachers to present open-ended questions in a gamified way.
Item selection	Puzzles whose solution requires participants to select one or more items (which can be either text or images) from a given set on one or more occasions. This resource allows teachers to present in a gamified way multiple-choice questions with single or multiple correct answers, as well as sorting questions.
Clock	Puzzles whose solution is a specific time of day consisting of hours, minutes, and, optionally, seconds.
Jigsaw	Puzzles whose solution requires participants to complete a jigsaw puzzle by placing each fragment of a given image in its correct position.
Conversation	Puzzles whose solution requires participants to enter a specific sequence of answers during a conversation. This resource allows teachers to present multiple-choice questions with a single correct answer in a gamified way.
Drag and drop classification	Puzzles whose solution requires participants to correctly classify a set of items represented by images into two categories.

2.8. Establishing an appropriate difficulty

Escapp allows teachers to modulate the difficulty of an educational escape room in several ways:

- Teachers can **create puzzles with solutions of varying lengths and formats**, which gives them flexibility in adjusting the difficulty of the puzzles. Moreover, for puzzles that must be solved by interacting with resources from the Escapp's resource catalog, teachers can generally adjust the difficulty by **changing the number of required interactions** through the available settings.
- Teachers can make puzzles easier or harder to solve by **changing the method used to verify participants' answers**. For example, they can make puzzles easier by using relaxed validations or wider numerical tolerances, or harder by requiring exact matches.
- Teachers can **increase or decrease the time available to participants to solve the puzzles** by changing the **duration** of the escape room in the **Basic Configuration** section.
- Teachers can **add clues and distractors** to the **team interface** through content blocks or scenes.
- Teachers can **use different approaches to delivering hints** to participants to help them solve the puzzles or disable hints altogether (see Section 2.9).
- Teachers can **specify the participation mode and team size** (see Section 2.3).

Moreover, teachers can **arrange puzzles in a sequence from easiest to hardest**, thereby implementing the recommended incremental difficulty approach.

2.9. Defining how to provide participants with timely help

Escapp allows teachers to **define hints and deliver them to participants using different approaches** in order to help them solve the puzzles while playing an educational escape room. With Escapp, it is possible to **define multiple hints for each escape room puzzle and specify their order** so that they can be delivered in increasing order of usefulness. Escapp also allows teachers to organize the hints for a puzzle into different **hint categories** and allow participants to choose the category from which they receive these hints. This feature allows participants to choose the task for which they need help when facing multi-task puzzles.

From the **Hint management** section, teachers can choose among three hint strategies: **disable hints**, **allow participants to obtain hints for free**, or **allow participants to obtain hints after successfully completing a quiz**. This latter strategy allows teachers to follow the recommendation in the IGLUE Best Practices guide that participants should incur a cost each time they request a hint. Teachers can also **limit the total number of hints** a participant can obtain, as well as the **frequency** with which hints can be obtained by specifying how long participants must wait before requesting a new hint.

Escapp also makes it easier for teachers to **deliver free hints to lagging participants or suggest that they request hints**. By default, when participants play an escape room in Escapp, they must request hints in order to receive them. However, teachers can enable **proactive hint delivery** so that hints are automatically sent to participants when they exceed the estimated time for solving a puzzle. Teachers can specify this time for each puzzle in the **Puzzles and hints** section. Alternatively, teachers can configure the escape room so that Escapp simply suggests that participants request a hint when the estimated time is exceeded. Moreover, Escapp makes it easier for teachers to identify lagging participants during the execution of an escape room through its **learning analytics dashboard** (see Section 4.5). Once these participants have been identified, teachers can opt to intervene to provide them with additional help. This help can be provided either by **sending real-time messages through Escapp** or in person without using the platform.

In addition to using pre-created hints, Escapp allows teachers to **provide participants with on-the-spot hints as a fallback** in supervised escape rooms when no more predefined hints are available. By default, once participants have obtained all the hints defined for an escape room puzzle, they are no longer allowed to request additional hints for that puzzle. However, teachers can enable **manual hints** for a specific escape room in the **Hint management** section. When this setting is enabled, once participants have exhausted the predefined hints for a puzzle, they can continue requesting hints and, when they obtain one, the platform will instruct them to ask the supervisors for it.

In order to **assist participants in the event of incidents or questions**, the IGLUE Best Practices guide recommends **supervising** educational escape rooms whenever possible. Escapp not only provides features that make it easier for teachers to help participants solve escape room puzzles while playing, but also provides features aimed at facilitating the supervision of educational escape rooms. These features are described in Section 4.4.

2.10. Defining the strategy of student grading

In the **Evaluation** section, teachers can **specify the criteria that will be used to calculate participants' grades** and **select how their participation in the escape room will be recorded**.

There are three options for **recording participation**. The first is not to record participation automatically. In this case, teachers can manually record in Escapp which of the users who enrolled in the escape room actually participated in the activity. The second option is to automatically mark as present all users who accessed the escape room while it was running. The third option is to automatically mark as present all users belonging to a team in which at least one member accessed the escape room while it was running. This last option can be particularly useful when there are not enough computers for all students and they need to share devices.

Regarding the **grading criteria**, teachers can assign a percentage of the grade to participation, as well as to each puzzle. In this way, teachers can **award students points simply for participating in the educational escape room**, as well as **grade them proportionally to their completion of the educational escape room** based on the puzzles they solve. If hints are enabled for an escape room, teachers also have the option to define a penalty or reward for each hint obtained, allowing them to consider additional factors when calculating grades.

Escapp automatically calculates the grades of all students who have participated in an educational escape room according to the grading criteria specified by the teachers. In the **Grading** section of the platform, teachers can view these grades and download them in CSV format, which allows them to easily import the grades into learning management systems such as Moodle. Furthermore, from the **learning analytics dashboard**, teachers can download detailed data related to the participants' interactions in CSV format. These data allow teachers to calculate grades based on multiple factors, such as the puzzles solved, the time spent solving each puzzle, the weight of each puzzle, and the hints obtained. In order to inform students about the grading criteria for an educational escape room, teachers can use the **Pre-activity instructions** section or the **description** field provided by Escapp.

2.11. Designing the escape room with accessibility and inclusivity in mind

Escapp provides several features that can help teachers **create and conduct inclusive and accessible educational escape rooms**:

- Escapp allows teachers to **create shifts that allow remote participation, either synchronous or asynchronous** (see Section 4.1). When there are barriers preventing participants from attending an educational escape room in person, teachers can make that escape room more **accessible** by offering shifts of this kind.
- In the **Publication** step, teachers can enable access to an educational escape room without signing up in Escapp, **allowing users to play the escape room without providing personal data**.
- Escapp has **multilingual support**, with its latest version supporting English, Spanish, and Serbian. Moreover, teachers can force the use of a specific language in the Escapp interface for participants in an escape room from the **Basic Configuration** section.
- Escapp allows teachers to **define clear instructions** using a rich text editor and to provide them to the participants of an educational escape room through the **initial page**, the **pre-activity instructions**, the **team interface**, and/or the **post-activity**

instructions. The text editor allows teachers to provide **clear, large, high-contrast text**, as well as **visual support for textual** content by incorporating images.

- Escapp allows teachers to upload and include **videos with subtitles**, as well as to provide **transcriptions for videos and audio files** using a rich text editor.
- Teachers can define different content blocks in the **team interface** in order to **present the same information in different formats**, supporting different learning styles and needs.
- The **cloning** feature offered by Escapp allows teachers to easily **create alternative versions of their escape rooms adapted for students with special needs**. These adaptations can include **additional time, extra hints, additional resources, or content presented in different formats**.
- Teachers can use Escapp to **monitor students with special needs more closely** through the **learning analytics dashboard**, as well as to **deliver free hints and additional instructions to them when necessary** using the **real-time messaging system**.

3. Build guidelines

3.1. Building the escape room with reusability in mind

Escapp supports the **reusability of educational escape rooms created with the platform** through several features:

- Escapp allows teachers to create and conduct **educational escape rooms composed exclusively of virtual resources** (i.e., **virtual educational escape rooms**), as well as **hybrid educational escape rooms that combine virtual and physical resources**.
- Escapp allows teachers to **run the same escape room an unlimited number of times**. The educational escape rooms are **reset automatically** by the platform before each new game. Teachers can define **multiple shifts** for different student groups, courses, or academic years.
- Teachers can **publish educational escape rooms under a Creative Commons license**, allowing others to reuse these resources.
- Teachers can **store all the resources** of an escape room in Escapp, allowing that escape room to function without relying on external resources. The virtual resources to be provided to participants through the **team interface** can be uploaded through the **resource catalog** of the escape room, whereas the physical resources needed to play a hybrid escape room can be provided by uploading a file in the **Pre-activity instructions** section.
- Escapp allows users to upload a file containing **instructions for teachers** for an educational escape room in the **Publication** section. This feature allows teachers to store in Escapp all the information necessary to properly conduct the escape room.
- Teachers can **store the source files** used to generate the content of an escape room (e.g., Word documents used to generate PDF files or Photoshop files used to generate images) by uploading them in a file together with the **instructions for teachers**.
- Teachers can invite other users to join the escape room as **collaborators** who can access and edit the escape room, thereby facilitating co-creation and shared ownership of the resource.
- Teachers can make **clones** of their escape rooms, which makes it easier for them to create new versions of an existing escape room, create new escape rooms using an existing one as a starting point, or run an existing escape room with new participants (e.g., a new class) while keeping the interaction data separate from data collected during previous runs.
- Teachers can create **backup copies** of an educational escape room by using the **export** button available on the editing page.

3.2. Using resources with low environmental impact

The IGLUE Best Practices guide states that a very effective and recommendable way to reduce the environmental impact of an educational escape room is to use virtual resources instead of physical ones. Escapp allows teachers to **create and conduct educational escape rooms composed exclusively of virtual resources** in which all puzzles, clues, hints, instructions, and feedback are delivered digitally through the web interface.

The interactive resources offered by Escapp through the **resource catalog** (e.g., keypad, decoder disk, hidden path, and jigsaw) can replace physical resources that would otherwise need to be purchased, crafted, or printed. In this regard, it is worth noting that the **Scene Maker** tool can remove the need for physical spaces by allowing teachers to create explorable digital environments that can be fictitious or represent real-world environments.

3.3. Considering changing the design of the escape room to make it easier to build

The IGLUE Best Practices guide recommends that teachers, whenever they encounter significant difficulties during the process of building an educational escape room, consider modifying the original design of the activity to simplify its construction. In this regard, it is worth pointing out that Escapp provides several features that **make it easier for teachers to modify an existing escape room** at any time:

- Teachers can change the **metadata and settings** of an educational escape room.
- Teachers can reorder the **puzzles** of an educational escape room, remove puzzles, add new puzzles, or modify the **hints** of a puzzle.
- Teachers can edit the **content blocks** previously added through the **Pre-activity instructions** section, the **Team interface** section, the **Classroom interface** section, or the **Post-activity instructions** section.
- Teachers can remove **uploaded resources**, as well as upload new ones.
- Teachers can update the **resources added through the Escapp's resource catalog** by modifying their settings.
- Teachers can modify the **scenes** created through the **Scene Maker** tool.
- Teachers can unpublish an escape room, returning it to draft status, and **republish** it after applying the desired changes.

4. Execution guidelines

4.1. Considering scheduling shifts and allowing asynchronous participation

Escapp allows teachers to **create shifts for an educational escape room**, which can be synchronous or asynchronous. **Synchronous shifts** are shifts that are scheduled for a specific date and can only be started by the author of the escape room or one of its collaborators, whereas **asynchronous shifts** are shifts that do not take place on a specific date but instead allow each participant to decide when to start playing. By default, users can enroll in a shift of an escape room at any time after the escape room is published. However, teachers can define an **enrollment period** for a shift, in which case users will only be able to enroll during that period. Moreover, teachers can restrict access to a shift with a **password** to ensure that only the intended users can access it, as well as **limit the number of participants** who can enroll in it. Teachers can create as many shifts as needed for an escape room, and each shift can have its own **name, type (synchronous or asynchronous), scheduled date (only for synchronous shifts), enrollment period, password, and capacity**. Moreover, teachers can conduct an escape room **face-to-face** in some shifts and **remotely** in others. Escapp allows teachers to **filter by shift** in several management sections, making it possible to view participants, teams, and analytics for specific shifts.

Thanks to the shifts feature, Escapp allows teachers to run educational escape rooms through **one or more synchronous sessions**, as well as to offer them as **asynchronous learning activities** that participants can start at any time within a set timeframe. By using shifts, teachers can facilitate the participation of users from different classes and time zones, as well as users with different needs.

Escapp also allows teachers to create **public educational escape rooms that participants can start and play at any time without the need to enroll in a shift and without any teacher intervention**. In order to create an escape room of this kind, teachers must select “Public” in the **visibility** field available in the **Publication** section. Teachers also have the option of allowing participants to restart and replay an escape room as many times as they wish by enabling the corresponding setting in the **Basic configuration** section.

4.2. Providing participants with initial instructions

Escapp has a **Pre-activity instructions** section that teachers can use to **provide participants with the initial instructions for an educational escape room**. Teachers can define these instructions using a **rich text editor**, which enables them not only to write text but also to embed multimedia resources such as **images, videos, and audios**, which can be uploaded to the **resource catalog** of the educational escape room. The possibility of embedding videos and audio allows teachers to **provide instructions in video or audio format**. Escapp also allows teachers to upload any type of file to the resource catalog and include links to these files in the initial instructions, allowing them, for instance, to provide the instructions as a PDF file. From the **Pre-activity instructions** section, teachers can also provide the physical resources that participants need to play a hybrid escape room by uploading a file that will be attached to the initial instructions.

The participants of an escape room will be able to access the instructions defined in the **Pre-activity instructions** section from the moment they enroll in the escape room. If teachers want participants to be able to access the instructions before enrolling, they can provide them in the **description** field available in the **Basic configuration** section.

4.3. Preparing the escape room for its execution

Escapp provides a **completion checklist** in the **Publication** section that makes it easier for teachers to check whether the escape rooms they create are ready to be conducted with the target audience. This checklist shows whether an escape room meets certain conditions, such as having all the mandatory information provided, having at least one puzzle properly defined, and having at least one hint defined if hints have been enabled. Furthermore, Escapp provides a set of **completion indicators** on the **editing page** of an escape room that visually indicate whether each setup section has been properly completed.

Before each execution of an educational escape room, it is recommended to conduct a quick test to ensure that everything works as intended. Escapp allows teachers to **test educational escape rooms** at any time (see Section 5.1).

Escapp **automatically resets** escape rooms before each new game. For **virtual escape rooms**, this reset should be sufficient to return the escape room to its initial state. However, for **hybrid escape rooms**, teachers may need to take additional actions with physical elements, as indicated in the IGLUE Best Practices guide. Once an escape room has been started, teachers can reset its state for an entire shift or, depending on the participation mode, for an individual team or participant. These resets can be useful for performing tests or solving problems during the execution of an escape room.

Escapp also makes it easier for teachers to **prepare new executions of escape rooms for which the collected data need to be stored separately**. Teachers can **clone** an existing escape room and use the clone to conduct a new execution in order to keep the collected data separate from the data collected during previous runs. Another way in which teachers can keep the data collected during a new execution of an escape room separate from previously collected data is by creating **new shifts**.

4.4. Supervising the escape room whenever feasible

Escapp facilitates the **supervision** of educational escape rooms through several features:

- In the **Basic configuration** section, teachers can specify a **link to a video conferencing room** (e.g., hosted on Zoom, Microsoft Teams, Google Meet, or Discord) for an escape room. Whenever this link is specified, it is provided to participants through a button located at the header of the **team interface**, allowing them to access the video conferencing room at any time during the game. This video conferencing room can be used by **supervisors** to assist participants who play **remotely** by answering questions, resolving technical issues, handling incidents and conflicts, or providing additional help with solving the puzzles. Escapp also allows this link to be provided together with the initial instructions for the escape room.
- The **real-time messaging** feature of Escapp allows the author of an escape room and its collaborators to **send messages to the participants in that escape room** while they are playing or waiting in the waiting room. It is possible to send a message to all participants in an escape room, to all participants in a specific shift, to all members of a team, or, in the case of single-player escape rooms, to an individual participant.

This feature enables **supervisors** to provide additional help and communicate instructions, clarifications, or encouragement without using external systems.

- Escapp provides a **learning analytics dashboard** that can be used to **monitor escape rooms in real time**. This dashboard provides real-time information about each participant's progress, allowing **supervisors** to easily identify participants who are struggling and may need assistance. See Section 4.5 for further details.
- On the one hand, the **Participants** section lists the users who are currently enrolled in the escape room and shows, for each user, their alias and email, the shift in which they are enrolled, their team (for escape rooms played in teams), and whether they have been marked as present. From this section, it is possible to filter participants by shift, cancel individual enrollments, manually mark participants as present (see Section 2.10 for information about automatic participation recording), and download the list of participants as a CSV file. On the other hand, the **Teams** section provides a real-time overview of all teams, including whether they have started playing, the shift to which they belong, and their current members. From this section, it is possible to filter teams by shift, send messages or emails to specific teams, restart their progress, or cancel their enrollment. For single-player escape rooms, this section treats each participant as a single-member team, thereby making it possible to perform the same actions for individual participants as those performed for teams. In summary, the **Participants** and **Teams** sections make it easier for **supervisors** to view enrollments, as well as check and track attendance for the escape room.
- Teachers can use the file that Escapp allows them to upload with **instructions for teachers** in order to provide **instructions for supervising the escape room**.
- Teachers can invite other users to join the escape room as **collaborators**. Since collaborators have access to all the information and content of the escape room (including puzzle solutions, hints, and instructions for teachers) and teachers can add as many collaborators as needed, this feature provides an easy way to **share with the supervisors all the information they need to properly supervise the escape room**. In this way, teachers can follow the recommendation in the IGLUE Best Practices guide that supervisors of an educational escape room should know all the details of the activity, including the solutions to all the puzzles. The collaborators of an escape room can also **edit settings, send real-time messages**, use the **learning analytics dashboard**, access the **Participants** and **Teams** sections, and perform all the actions available in these sections. Therefore, the collaborators feature also enables **supervisors** to use all the tools that Escapp provides to conduct and supervise educational escape rooms.

Escapp also allows teachers to **conduct unsupervised educational escape rooms that participants can start and play without their intervention** (see Section 4.1). The IGLUE Best Practices guide recommends using automated hints and providing final hints with the puzzle solutions in educational escape rooms intended to be conducted without supervision. As described in Section 2.9, Escapp provides teachers with all the features they need to easily follow this recommendation.

4.5. Monitoring participants while playing the escape room

Whenever an educational escape room is conducted through Escapp, the platform **automatically monitors** the activity in order to **track the progress and performance of each participant**. Furthermore, if automatic participation recording has been enabled (see Section 2.10), Escapp will also automatically track **each participant's attendance**. This attendance can also be tracked manually by the supervisors.

Escapp collects all the data that the IGLUE Best Practices guide recommends collecting as part of the monitoring process. For every educational escape room, Escapp automatically collects, for each team or, in the case of single-player escape rooms, each participant, a **record indicating whether they accessed the escape room through Escapp, the puzzles solved and the time at which each puzzle was solved, the failed attempts to solve puzzles and the time at which each of these attempts was made, the hints delivered and the time at which each hint was delivered**, and, in the case of escape rooms in which participants are required to successfully complete a quiz to obtain a hint, the **failed attempts to obtain hints and the time at which each of these attempts was made**. Furthermore, Escapp stores the **identifiers (aliases or real names)** of all participants, a **record indicating which participants attended the activity** (which can be filled in manually or automatically), and the **members of each team**.

The **Participants** section shows the users who enrolled in the escape room, indicating their identifiers and which users have been automatically or manually marked as present. The **Teams** section shows the teams enrolled in the escape room and specifies their members. This section also indicates which teams have started playing (i.e., which teams have accessed the escape room through Escapp) or, in the case of single-player escape rooms, which participants have started playing. See Section 4.4 for details about the **Participants** and **Teams** sections.

Escapp provides a **learning analytics dashboard** that presents **various data automatically obtained from participant interactions**, which can be used by supervisors to **monitor an educational escape room in real time**. By using this dashboard, the author of an escape room and its collaborators can view:

- A **quick snapshot** of the escape room with the following indicators: total number of participants, success rate (i.e., the percentage of teams/participants that solved all the escape room puzzles in time), best completion time, average team size, average number of hints obtained per team/participant, number of teams/participants that solved each puzzle, and number of teams/participants that obtained each hint.
- A **leaderboard** including all teams or participants in the case of single-player escape rooms.
- A table showing how much **time** each team/participant took to complete each puzzle.
- A table showing aggregated data for each puzzle, including the **average, standard deviation, minimum, and maximum for completion time** and the **number of failed attempts**.
- A graph representing how teams/participants **progressed** through the escape room by solving puzzles over time.

- A **timeline** visually showing the experience of each team/participant over time, with colored bars representing the time spent on each puzzle and markers representing the start and completion of the escape room, the failed and successful attempts to solve puzzles, the hints obtained, and the failed attempts to obtain hints.
- A graph showing **how many hints were obtained** by each team/participant and, for escape rooms in which participants are required to successfully complete a quiz to obtain a hint, **how many failed attempts to obtain hints were made** by each team/participant.
- A graph showing the **distribution of puzzle completions over time**, indicating how many teams/participants solved each puzzle at any given time.

The **learning analytics dashboard** allows supervisors to **filter data by shift**, which makes it easier for them to monitor the group of participants currently playing the escape room. During a live session, supervisors can use the **timeline** provided by the **learning analytics dashboard** to easily **identify participants who are struggling or falling behind and who may need extra help**. The timeline also makes it easier for supervisors to know when participants solve new puzzles and thus to provide them with **positive reinforcement**.

4.6. Helping participants to solve the puzzles when necessary

Section 2.9 describes in detail how Escapp allows teachers to **define hints and automatically deliver them to participants using different approaches** in order to help them solve the puzzles while playing an educational escape room. When **automated hints** are used in an escape room, it is highly recommended to explain to participants how to request and obtain these hints in the **initial instructions**. Section 4.2 describes how teachers can provide participants with these instructions when conducting an educational escape room using Escapp.

As indicated in the IGLUE Best Practices guide, even if the **planned hint approach** is adequate, certain circumstances may require small adjustments during the course of an educational escape room. Next, this section describes Escapp's features that can help **supervisors** make adjustments of this kind.

During an educational escape room, Escapp allows supervisors to identify **lagging participants** using its **learning analytics dashboard** (see Section 4.5) and send them, through **real-time messages**, **customized free hints to help them solve the puzzle they are currently facing**. Instead of providing participants with free hints, supervisors can also simply send them **messages advising them to request a hint**. Escapp also allows supervisors to simultaneously provide **free hints** to all participants who are currently playing the escape room by **sending real-time messages that are broadcast to all of them**. This feature can be useful for **easily providing collective assistance** when a significant number of participants are at risk of not completing the escape room or when many participants are stuck on the same puzzle. One of the measures suggested by the IGLUE Best Practices guide to help participants solve the puzzles under certain circumstances is to **provide an additional hint each time a participant obtains one**. Supervisors can view the hints obtained by each team/participant in the **learning analytics dashboard** and provide additional hints to them by sending real-time messages. Thus, supervisors can adopt this measure when using Escapp.

4.7. Motivating participants while playing the escape room

The IGLUE Best Practices guide proposes various measures that teachers can take to **boost participant motivation throughout an educational escape room**. This section describes the Escapp features that make it easier for teachers to apply these measures.

- Escapp allows teachers to add a **countdown timer** to an educational escape room in order to show participants at all times how much time they have left to complete the escape room. Teachers can add a countdown timer to both the **team interface** (allowing participants to see it on their own devices) and the **classroom interface** (allowing it to be projected in a classroom or other location). Escapp also allows teachers to give participants periodic updates on the remaining time by **broadcasting real-time messages**.
- Escapp allows teachers to add a **progress bar** to an educational escape room in order to show each team or, in the case of single-player escape rooms, each participant the percentage of escape room puzzles that they have solved. In this way, all players can see at all times their **progress toward completing the escape room**. The **progress bar** can only be added to the **team interface**, since its value is specific to each team or participant.
- Escapp allows teachers to add a **leaderboard** to an educational escape room, both through the **team interface** and the **classroom interface**. In single-player escape rooms, the leaderboard includes all participants in the session and displays each participant's alias. In contrast, in escape rooms that allow participants to play in teams, the leaderboard includes all teams in the session and displays each team's name. For each participant/team, the leaderboard indicates the number of puzzles solved and, if they have completed the escape room, the total time taken to do so. The ranking is determined based on two criteria: participants/teams that have solved a greater number of puzzles are ranked higher, and, in the event of a tie, those that have taken less time to complete the escape room are ranked higher. During the execution of the escape room, the leaderboard is **automatically updated** in real time.
- On the one hand, the **success messages** that teachers can define for the puzzles can be used to provide **positive reinforcement** to participants (e.g., by congratulating them on solving a puzzle or offering encouraging words). On the other hand, the **failure messages** that teachers can define for the puzzles can be used to **praise participants for their efforts and encourage them to keep trying**.
- Escapp allows **supervisors** to **send real-time messages** to participants in order to **motivate them or encourage collaboration and communication**. These messages can be sent to all participants in a session or targeted at a specific team/participant.
- Teachers can view all the participants who played the escape room in the **Participants** section and use the **leaderboard** or the data provided by the **learning analytics dashboard** to easily identify the best-performing participants or simply all those who completed the escape room. In this way, Escapp makes it easier for teachers to **offer rewards or certificates** to these participants. Furthermore, teachers can use the **Post-activity instructions** section to provide all participants with a final reward or give them instructions for obtaining one.

4.8. Conducting a debriefing session

Some of the features of Escapp can help teachers conduct a **debriefing session** after participants finish playing an educational escape room.

Teachers can use the **Post-activity instructions** section to provide participants with resources for the debriefing session, such as a set of questions to stimulate reflection, critical thinking, and discussion, the correct answers and explanations of those answers, a summary of the skills put into practice and their relevance, information on how the escape room relates to broader learning processes as well as to professional activities and everyday situations, a link to a survey to collect opinions, reflections, and other information about the experience, and links to additional learning resources to delve deeper into the topics addressed by the escape room.

Moreover, during a debriefing session, teachers can show participants the **learning analytics dashboard** provided by Escapp in order to allow them to view various data derived from the information about their interactions automatically collected during the activity. The data available in this dashboard are described in detail in Section 4.5. To show participants the **learning analytics dashboard** in educational escape rooms conducted face-to-face, teachers can project it in the room where the activity took place, whereas in educational escape rooms conducted remotely, they can share their screen using a video conferencing system. The data presented by the **learning analytics dashboard**, such as the **puzzle statistics**, the **distribution of puzzle completions over time**, the **progress graph**, and the **timeline**, can facilitate discussion about which puzzles were the most and least challenging, how time was managed, and which of the strategies adopted worked best.

5. Evaluation guidelines

5.1. Testing the escape room thoroughly before its execution

Escapp allows the author of an educational escape room and any of its collaborators to **play the escape room as if they were a participant in order to test it**. In these **test runs**, the escape room can be **tested individually or as a group**, in which case all those taking part in the test run will play on the same team. Teachers can conduct test runs **as many times as they wish and at any time**, even before publishing the escape room (i.e., while it is still in draft status). Therefore, they can test escape rooms using Escapp before executing them. Teachers can also edit an escape room while testing it, which can be useful for correcting errors, adding new content, or making adjustments.

For **virtual escape rooms**, Escapp allows teachers to perform all the checks needed to ensure that the escape room will work as intended. However, for **hybrid escape rooms**, teachers will need to perform additional checks in order to ensure that the physical elements work as intended.

Next, we include **a list of checks that teachers can perform when conducting a test run with Escapp**:

- Verify that the instructions provided to participants through Escapp contain all the necessary information and are clear and easy to understand.
- Verify that all files (e.g., images, videos, and audios) load correctly, that all virtual content is correct, and that all virtual resources (including scenes created using Scene Maker and resources linked to puzzles such as keypads and decoder disks) work as intended.
- Verify that all puzzles can be solved as expected, that the solution to each puzzle can be figured out using the information and resources available at the time the puzzle is presented, and that appropriate feedback is provided when each puzzle is solved.
- Verify that the messages displayed when participants fail to solve puzzles are correct.
- Verify that each content block is shown during the expected stages and, if scenes are used, verify that each of their screens is accessible during the expected stages.
- Verify that the time available for participants to solve all the puzzles is as intended.
- Review the hints for each puzzle, check that they are helpful in progressing through the escape room, and verify that they are delivered as intended.

In addition to conducting test runs, **teachers can test their educational escape rooms through Escapp before executing them with their intended audience by conducting pilot runs, which require using student accounts**. To conduct a pilot run of an escape room with Escapp, a teacher should publish the escape room with **unlisted** visibility and create a **shift with password-restricted access**. Then, the participants in the pilot run should enroll in the escape room using an Escapp student account and the specified password. Lastly, the escape room should be started by the teacher. Pilot runs allow teachers to test educational escape rooms with users who have not been involved in their creation (e.g., volunteer students), who can participate **individually or in multiple teams**.

5.2. Testing the escape room from the early stages of development

As explained in Section 5.1, Escapp allows teachers to **test educational escape rooms by conducting test runs at any time, even while the escape rooms are in draft status**. Therefore, teachers can test their escape rooms through Escapp while they are still **under development**.

Moreover, Escapp allows teachers to test a whole educational escape room and each of its puzzles by using **virtual prototypes**. Teachers can create **digital mockups** (e.g., in the form of text, images, or simple web applications) and include them in an escape room that is being developed by writing text, uploading files, and embedding resources. By using Escapp, teachers can easily define a set of preliminary puzzles, specify which mockups should be displayed for each of them, and conduct a test run to verify that each of these puzzles works as intended. Escapp also allows teachers to **preview the resources added using the Escapp's resource catalog**, including resources linked to puzzles and scenes created through the **Scene Maker** tool, so that they can test those resources individually at any time.

The test runs that Escapp allows teachers to conduct can be used for **testing technology and digital tools through proofs of concept**. In this way, teachers can perform tests in order to select the technology and digital tools that they consider most appropriate for building the resources of an educational escape room.

Escapp also has some features that can make it easier for teachers to **share or show a preliminary version of an educational escape room with other teachers or students in order to get early feedback**. On the one hand, thanks to **Escapp's collaborators feature**, a teacher can share an escape room with other teachers at any time, allowing them to play the escape room and access all its resources and information, including instructions, learning objectives, settings, hints, and puzzle solutions. On the other hand, teachers can conduct **pilot runs** through Escapp (see Section 5.1) in order to **test preliminary versions of an educational escape room with students or other users who have not been involved in its creation**.

5.3. Evaluating the impact of the escape room on participants' perceptions and learning

Escapp has some features that can make it easier for teachers to assess the **participants' perceptions of an educational escape room and its impact on their learning**:

- Teachers can use the **Post-activity instructions** section to **provide participants with a link to a questionnaire for collecting their perceptions of the escape room after they finish playing**.
- Teachers can **provide participants with a link to a pre-test before playing and a link to a post-test after playing** using the **Pre-activity instructions** and the **Post-activity instructions** sections, respectively. By using a pre-test and a post-test, teachers can measure the learning gains experienced by the participants as a result of participating in the escape room.
- Escapp **automatically collects information about the participants' interactions** during the execution of educational escape rooms and provides a **learning analytics dashboard** that allows teachers to **view various data derived from this information mainly through graphs and tables**. Section 4.5 describes in detail which data are

displayed by this dashboard. From the **learning analytics dashboard**, in addition to viewing data, teachers can **download, in CSV format, various data automatically generated or collected by Escapp that are related to the participants' interactions**. This feature enables teachers and researchers to view, process, and analyze the data using external tools, for example, to perform more in-depth statistical analyses, calculate grades based on participants' performance, create custom visualizations, compare the performance of different groups, identify patterns in participants' behavior, or combine the data with data from other sources. The data, graphs, and tables that the **learning analytics dashboard** provides to teachers can help them **understand how participants behave during an educational escape room** as well as **improve the escape room for future editions**.

